HONORS CHEMISTRY • UNIT 4

Chemical Reactions

Write, balance, classify and predict the products of chemical reactions, then show only the particles that actually change.

TEACHER EDITION

Everything in a light-gray box or cell is an answer or model response. Teacher notes (misconceptions, demo ideas, discussion prompts, sources) appear in the heavy-bordered boxes labeled TEACHER NOTE. The pages follow the student packet, so they line up closely.

What we will learn

4.1	Balancing equations & writing reactions
4.2	Types of reactions
4.3	Predicting products: synthesis, decomposition, combustion
4.4	Predicting products: single replacement (activity series)
4.5	Predicting products: double replacement (solubility rules)
4.6	Net ionic equations
4.7	Naming & writing formulas (review)

By the end of this unit I can...

- Balance equations using coefficients and explain why equations must be balanced.
- Translate between word equations and balanced symbol equations with state symbols.
- Classify a reaction as synthesis, decomposition, single replacement, double replacement or combustion.
- Predict the products of synthesis, decomposition and combustion reactions.
- Use the activity series to predict single replacement reactions and solubility rules to predict precipitates.
- Write complete, ionic and net ionic equations and identify spectator ions.
- Name and write formulas for ionic compounds, molecular compounds, acids and polyatomic ions.

TEACHER OVERVIEW

Numbering: the homework folder calls this "Unit 4 – Reactions" (topics 4.1–4.7). This packet uses the same numbers: Unit 4, topics 4.1–4.7, in the same order.

Answers are computed. Balanced equations, double-replacement products, solubility and net ionic equations in the key were generated by code (atoms and charges checked), then proofread.

Homework quirks fixed in the packet: HW 4.2 #5 is written $2La_2O_3 \rightarrow 4Al + 3O_2$ (typo; Al_2O_3 used) and #7 is $4C_4H_9O + 27O_2$ (not a real compound; butanol, $C_4H_{10}O$, used). HW 4.5 #13 lists $BaSO_4$ as aqueous (it is insoluble); the key says no reaction. HW 4.4 lists many reactants as solids; the key uses the same states as the homework only where meaningful.

Misconceptions: changing subscripts to balance; forgetting diatomic elements (H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2); writing weak electrolytes/solids/gases as ions in net ionic equations; assuming every double replacement "reacts" (no precipitate/gas/water = no reaction); mixing up the activity series direction (a metal replaces a LESS active metal's ion).

4.1 Balancing Chemical Equations & Writing Reactions

Law of conservation of mass • coefficients • state symbols

BIG IDEA

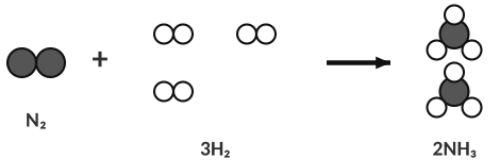
Atoms are not created or destroyed in a chemical reaction, so every atom on the **reactant** side must appear on the **product** side. We balance by changing **coefficients** (the big numbers in front), never the subscripts.

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Reactant / product	Reactants are the starting substances (left of the arrow); products are the new substances formed (right).	$H_2 + O_2 \rightarrow H_2O$
Coefficient vs subscript	A coefficient tells how many units of a substance take part; a subscript tells how many atoms are in one unit and can never be changed (it would change the substance).	$2H_2O$: 2 is the coefficient, ₂ the subscript
Law of conservation of mass	Matter is neither created nor destroyed in a chemical reaction; the mass and the number of each kind of atom stay the same.	4 H atoms in = 4 H atoms out
State symbols	(s) solid, (l) liquid, (g) gas, (aq) aqueous = dissolved in water.	NaCl(aq)

1. Why does a chemical equation need to be balanced? What does a balanced equation represent? (Think about a cooking recipe.)

Because of the law of conservation of mass: atoms can't disappear or appear, so the same atoms must be on both sides. A balanced equation is like a recipe: it shows the exact ratio of particles (and moles) of each reactant needed and each product made, e.g. $2\ H_2 + O_2 \rightarrow 2\ H_2O$ means 2 molecules (or moles) of hydrogen react with 1 of oxygen to make 2 of water.



$N_2 + 3H_2 \rightarrow 2NH_3$: 2 N and 6 H on each side.

HOW TO BALANCE (TRY THIS ORDER)

1. Write the correct formulas (diatomic elements: **H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2**).
2. Make an atom tally for each side. Treat a polyatomic ion (PO_4, NO_3, SO_4 …) as one unit if it stays together.
3. Balance metals first, then non-metals, then **H, then O last**. Elements that appear in one compound on each side are easiest.
4. If you get a fraction, multiply everything by 2. Re-check every atom. Coefficients must be the lowest whole numbers.

WORKED EXAMPLE — A COMBUSTION

Balance: ___ C_2H_6 + ___ O_2 → ___ CO_2 + ___ H_2O

C: 2 on the left → put 2 in front of CO_2. H: 6 on the left → 3 H_2O. O on the right: 2(2) + 3 = 7 → needs 3½ O_2. Multiply all by 2: $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$ ✓ C 4/4, H 12/12, O 14/14

Practice: balance each equation

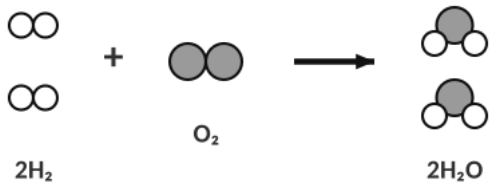
Use the space under each equation for your atom tally (count each element on both sides) and your trial coefficients.

	Balance each equation. Show your atom tally and work below it, then write the final balanced equation.
a.	___ Na + ___ O_2 → ___ Na_2O ANSWER: $4Na + O_2 \rightarrow 2Na_2O$
b.	___ K + ___ B_2O_3 → ___ K_2O + ___ B ANSWER: $6K + B_2O_3 \rightarrow 3K_2O + 2B$
c.	___ H_3PO_4 + ___ KOH → ___ K_3PO_4 + ___ H_2O ANSWER: $H_3PO_4 + 3KOH \rightarrow K_3PO_4 + 3H_2O$

	Balance each equation. Show your atom tally and work below it, then write the final balanced equation.
d.	___ Al + ___ S_8 → ___ Al_2S_3 ANSWER: $16Al + 3S_8 \rightarrow 8Al_2S_3$
e.	___ N_2 + ___ O_2 → ___ N_2O_5 ANSWER: $2N_2 + 5O_2 \rightarrow 2N_2O_5$
f.	___ H_3PO_4 + ___ $Mg(OH)_2$ → ___ $Mg_3(PO_4)_2$ + ___ H_2O ANSWER: $2H_3PO_4 + 3Mg(OH)_2 \rightarrow Mg_3(PO_4)_2 + 6H_2O$
g.	___ NH_3 + ___ O_2 → ___ N_2 + ___ H_2O ANSWER: $4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$
h.	___ FeS_2 + ___ O_2 → ___ Fe_2O_3 + ___ SO_2 ANSWER: $4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$

2. Draw particle diagrams to balance $H_2 + O_2 \rightarrow H_2O$. Draw the correct number of molecules on each side.

$2\ H_2 + O_2 \rightarrow 2\ H_2O$ (4 H and 2 O on each side)



Answer

Writing equations from words

HELPFUL REMINDERS

- Diatomic elements are always written as molecules: H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2.
- Use the ion charges to write correct formulas (criss-cross) and then balance.
- Add states: (s), (l), (g), (aq).

Polyatomic ion	Name	Polyatomic ion	Name
NH_4^+	ammonium	SO_4^{2-}	sulfate
NO_3^-	nitrate	CO_3^{2-}	carbonate
OH^-	hydroxide	PO_4^{3-}	phosphate
$C_2H_3O_2^-$	acetate	ClO_3^-	chlorate
MnO_4^-	permanganate	CN^-	cyanide

Reaction described in words	Balanced equation with states
Sodium metal reacts with chlorine gas to produce sodium chloride.	$2Na\ (s) + Cl_2\ (g) \rightarrow 2NaCl\ (s)$
Aluminum metal reacts with aqueous copper(II) nitrate to produce aqueous aluminum nitrate and copper metal.	$2Al\ (s) + 3Cu(NO_3)_2\ (aq) \rightarrow 2Al(NO_3)_3\ (aq) + 3Cu\ (s)$
Aqueous nitric acid reacts with aqueous magnesium hydroxide to produce aqueous magnesium nitrate and liquid water.	$2HNO_3\ (aq) + Mg(OH)_2\ (aq) \rightarrow Mg(NO_3)_2\ (aq) + 2H_2O\ (l)$
Propane gas (C_3H_8) reacts with oxygen gas to produce gaseous carbon dioxide and gaseous water.	$C_3H_8\ (g) + 5O_2\ (g) \rightarrow 3CO_2\ (g) + 4H_2O\ (g)$
Solid aluminum oxide decomposes into solid aluminum and oxygen gas.	$2Al_2O_3\ (s) \rightarrow 4Al\ (s) + 3O_2\ (g)$
Solid ammonium chloride decomposes into gaseous nitrogen trihydride and gaseous hydrochloric acid.	$NH_4Cl\ (s) \rightarrow NH_3\ (g) + HCl\ (g)$
Aqueous lead(II) nitrate and aqueous potassium iodide react to produce solid lead(II) iodide and aqueous potassium nitrate.	$Pb(NO_3)_2\ (aq) + 2KI\ (aq) \rightarrow PbI_2\ (s) + 2KNO_3\ (aq)$

Turn each equation into a word equation:

Equation	Word equation
$Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq)$	Solid sodium oxide reacts with liquid water to produce aqueous sodium hydroxide.
$CS_2(l) + 3O_2(g) \rightarrow CO_2(g) + 2SO_2(g)$	Liquid carbon disulfide reacts with oxygen gas to produce carbon dioxide gas and sulfur dioxide gas.
$K(s) + CuNO_3(aq) \rightarrow KNO_3(aq) + Cu(s)$	Solid potassium reacts with aqueous copper(I) nitrate to produce aqueous potassium nitrate and solid copper.
$2Li(s) + 2H_2O(l) \rightarrow 2LiOH(aq) + H_2(g)$	Solid lithium reacts with liquid water to produce aqueous lithium hydroxide and hydrogen gas.

TEACHER NOTE

Note: the K + $CuNO_3$ line is as written in the homework (copper(I) nitrate); it is balanced as K + $CuNO_3 \rightarrow KNO_3$ + Cu.

4.2 Types of Reactions *Six patterns*

Type	General pattern	Clue	Example
Synthesis	A + B → AB	2 or more reactants, 1 product	$2Na + Cl_2 \rightarrow 2NaCl$
Decomposition	AB → A + B	1 reactant (a binary compound), breaks into its elements	$2H_2O \rightarrow 2H_2 + O_2$
Single replacement	A + BC → AC + B	element + compound → new element + new compound	$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
Double replacement	AB + CD → AD + CB	two compounds swap partners (precipitate or gas)	$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
Neutralization	HA + MOH → MA + H_2O	an acid plus a base makes a salt and water	$HCl + NaOH \rightarrow NaCl + H_2O$
Combustion	$C_xH_y + O_2 \rightarrow CO_2 + H_2O$	hydrocarbon (or similar) + O_2, heat and light	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Acids and bases: how do I tell them apart?

Key terms — write each definition in your own words

Term	Definition (your words)	Example / sketch
Acid	A compound that releases hydrogen ions (H^+) in water. The formula usually starts with H.	HCl, H_2SO_4, HNO_3, $HC_2H_3O_2$
Base	A compound that releases hydroxide ions (OH^-) in water. The formula usually ends with OH (metal + OH).	NaOH, $Ca(OH)_2$, KOH
Salt	An ionic compound made of the metal from the base and the non-metal part of the acid.	NaCl from HCl + NaOH

NEUTRALIZATION

An acid and a base swap partners: the H^+ from the acid joins the OH^- from the base to make **water**, and the leftover ions form a **salt**. **acid + base → salt + water**

Predict the salt and water

Reactants	Balanced equation	Name of the salt
HCl + NaOH	$HCl + NaOH \rightarrow NaCl + H_2O$	sodium chloride
H_2SO_4 + KOH	$H_2SO_4 + 2KOH \rightarrow K_2SO_4 + 2H_2O$	potassium sulfate
HNO_3 + $Ca(OH)_2$	$2HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + 2H_2O$	calcium nitrate
HBr + $Al(OH)_3$	$3HBr + Al(OH)_3 \rightarrow AlBr_3 + 3H_2O$	aluminum bromide

TEACHER NOTE

Neutralization is really a double replacement that makes water, so students may also call it that; accept either on early quizzes but the packet asks for "neutralization" whenever an acid and a base react. Classic demos: universal indicator in HCl then add NaOH; or magnesium hydroxide suspension dropped into dilute HCl with indicator.

Classify each reaction

#	Reaction	Type
1	$C_5H_{12} + 8O_2 \rightarrow 5CO_2 + 6H_2O$	combustion
2	$2K_3PO_4 + 3BaCl_2 \rightarrow Ba_3(PO_4)_2 + 6KCl$	double replacement
3	$2Al(OH)_3 + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 6H_2O$	neutralization
4	$2AgNO_3 + Na_2CO_3 \rightarrow Ag_2CO_3 + 2NaNO_3$	double replacement
5	$2Al_2O_3 \rightarrow 4Al + 3O_2$	decomposition
6	$2Al + 6HCl \rightarrow 3H_2 + 2AlCl_3$	single replacement
7	$C_4H_{10}O + 6O_2 \rightarrow 4CO_2 + 5H_2O$	combustion
8	$3HBr + Al(OH)_3 \rightarrow 3H_2O + AlBr_3$	neutralization
9	$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$	synthesis
10	$2HgO \rightarrow 2Hg + O_2$	decomposition
11	$Na_2O + H_2O \rightarrow 2NaOH$	synthesis
12	$Cu + 2AgNO_3 \rightarrow 2Ag + Cu(NO_3)_2$	single replacement

TEACHER NOTE — CORRECTIONS

Equations 3 and 5 have been corrected from the homework ($Al(OH)_3 + H_2SO_4$ gives 6 H_2O when balanced; Al_2O_3 replaces the typo La_2O_3) and 7 uses butanol in place of the non-existent C_4H_9O. In #9, $NH_3 + H_2SO_4$ is a synthesis as written (a base without OH^- in its formula); mention it if students ask.

1. Which reaction type has the pattern AB → A + B?

A) synthesis **B) decomposition** C) single replacement D) combustion

2. $HNO_3 + KOH \rightarrow KNO_3 + H_2O$ is…

A) synthesis B) decomposition **C) neutralization** D) combustion

4.3 Predicting Products: Synthesis, Decomposition & Combustion

Reactants → products

PREDICTION RULES

Synthesis (two elements combine): a **metal + non-metal** makes an ionic compound. Use the group number to find each charge (Group 1 → 1+, Group 2 → 2+, Group 3 → 3+; Group 15 → 3−, Group 16 → 2−, Group 17 → 1−), then criss-cross the charges to write the formula ($Mg^{2+} + O^{2-} \rightarrow MgO$). (If a problem combines two non-metals, the product's formula will be given.)

Decomposition: a binary compound (two elements) breaks apart into its two **pure elements** (by electrolysis or strong heat). Remember the diatomic elements: H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2.

Combustion: hydrocarbon (or an alcohol) + $O_2 \rightarrow CO_2 + H_2O$ (complete combustion).

Complete and balance

	Reactants →	Balanced equation (products and coefficients)	Type
1	$Na + Cl_2 \rightarrow$	$2Na + Cl_2 \rightarrow 2NaCl$	synthesis
2	$Mg + O_2 \rightarrow$	$2Mg + O_2 \rightarrow 2MgO$	synthesis
3	$Al + S_8 \rightarrow$	$16Al + 3S_8 \rightarrow 8Al_2S_3$	synthesis
4	$Li + N_2 \rightarrow$	$6Li + N_2 \rightarrow 2Li_3N$	synthesis
5	$NaCl \rightarrow$	$2NaCl \rightarrow 2Na + Cl_2$	decomposition

	Reactants →	Balanced equation (products and coefficients)	Type
6	$H_2O \rightarrow$	$2H_2O \rightarrow 2H_2 + O_2$	decomposition
7	$Al_2O_3 \rightarrow$	$2Al_2O_3 \rightarrow 4Al + 3O_2$	decomposition
8	$CH_4 + O_2 \rightarrow$	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	combustion
9	$C_2H_6 + O_2 \rightarrow$	$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$	combustion
10	$C_2H_4 + O_2 \rightarrow$	$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$	combustion

TEACHER NOTE

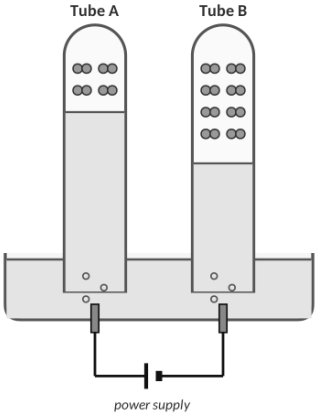
Items 1–3: students find charges from the group number and criss-cross (Na^+Cl^-, $Mg^{2+}O^{2-}$, $Al^{3+}S^{2-} \rightarrow Al_2S_3$). Item 4 ($Li_3N$) uses the same method with a 3– nitride ion. Non-metal + non-metal syntheses are left out because their products can't be predicted from charges; if you include one, give the formula. Iron and other transition metals are left out of synthesis prediction because their charge isn't fixed. Electrolysis of molten $NaCl/Al_2O_3$ is not a bench demo; show a video. Incomplete combustion (CO or C) can be mentioned but is not required.

1. Propane (C_3H_8) burns completely in oxygen. Which products form?

A) CO and H_2 **B) CO_2 and H_2O** C) C and H_2O D) C_3H_8 and O_2

LAB CORNER • Electrolysis of water

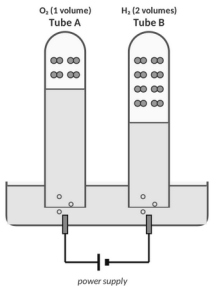
A power supply is connected to two electrodes in water. Gas collects in two inverted tubes in a volume ratio of 2 : 1.



Each pair of dots is one gas molecule. Both tubes were filled by the same power supply for the same time.

2. Look at the amounts of gas collected. (a) Which tube holds hydrogen and which holds oxygen? Explain how you decided. (b) Write the balanced decomposition equation for water.

(a) Tube B has twice as much gas as Tube A, so Tube B is hydrogen and Tube A is oxygen. The balanced equation is $2H_2O\ (l) \rightarrow 2H_2\ (g) + O_2\ (g)$; the coefficients 2 : 1 mean twice as many molecules of H_2 as O_2, and equal numbers of gas molecules take up equal volumes.
(b) $2H_2O\ (l) \rightarrow 2H_2\ (g) + O_2\ (g)$



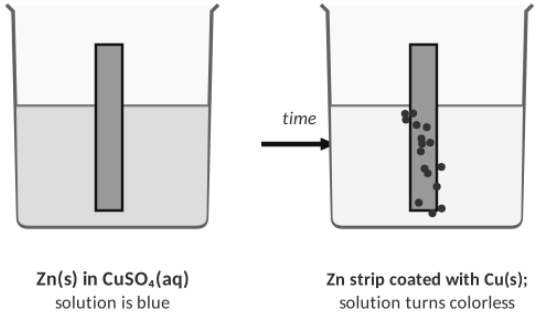
Answer key: gases labeled.

4.4 Single Replacement & the Activity Series

A more active element replaces a less active one

BIG IDEA

In a single replacement reaction a free element swaps places with an element in a compound. It only works if the free element is **more active** (more eager to lose or gain electrons) than the one it replaces.



Zinc is more active than copper: $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$.

Activity series	**Most active → least active**
Metals	Li, K, Ba, Sr, Ca, Na, Mg, Al, Zn, Fe, Ni, Pb, (H), Cu, Ag, Au
Halogens	$F_2 > Cl_2 > Br_2 > I_2$
Reacts with cold water?	Li, K, Ba, Sr, Ca, Na → metal hydroxide + H_2 ($2Na + 2H_2O \rightarrow 2NaOH + H_2$); metals below Ca, Na do not.

1. Explain in words how to decide whether a single replacement reaction will happen.

Find the free element and the element it would replace in the compound in the activity series. If the free element is higher (more active) than the element in the compound, it replaces it; if it is lower, there is no reaction (NR). For halogens, compare halogens with the anion.

2. Write the balanced equation for $Zn + CuSO_4$ and tell which metal is more active and how you know.

$Zn (s) + CuSO_4 (aq) \rightarrow ZnSO_4 (aq) + Cu (s)$. Zinc is more active than copper: it replaced the copper ion in the compound, and the copper metal came out of solution.

Predict the products (write NR if no reaction)

	Reactants	**Products (balanced) or NR**
1	$Li + BaCl_2 \rightarrow$	$2Li + BaCl_2 \rightarrow 2LiCl + Ba$
2	$Li + CaS \rightarrow$	$2Li + CaS \rightarrow Li_2S + Ca$
3	$K + BaCl_2 \rightarrow$	$2K + BaCl_2 \rightarrow 2KCl + Ba$
4	$Ba + NaF \rightarrow$	$Ba + 2NaF \rightarrow BaF_2 + 2Na$
5	$Na + MgCl_2 \rightarrow$	$2Na + MgCl_2 \rightarrow 2NaCl + Mg$

	Reactants	Products (balanced) or NR
6	$F_2 + NaCl \rightarrow$	$F_2 + 2NaCl \rightarrow 2NaF + Cl_2$
7	$Cl_2 + NaF \rightarrow$	NR (Cl_2 is less active than F_2)
8	$Cu + FeSO_4 \rightarrow$	NR (Cu is less active than Fe)
9	$Na + H_2O \rightarrow$	$2Na + 2H_2O \rightarrow 2NaOH + H_2$
10	$Ag + H_2O \rightarrow$	NR (Ag is far below Na in the series)

TEACHER NOTE

The homework lists several of these reactants as (s), (aq) etc. inconsistently (e.g., $BaCl_2(s)$); the packet leaves states out for the prediction task. Compare Na and Mg carefully: Na is more active than Mg in the standard series, so Na + $MgCl_2$ does react.

3. Which will NOT react?

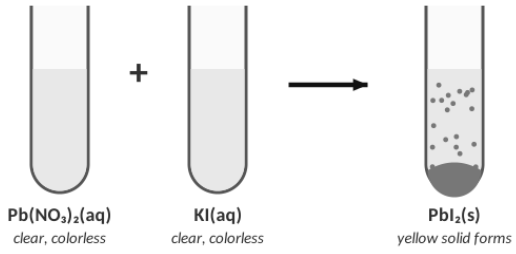
A) $Zn + CuSO_4$ B) $Mg + FeCl_2$ **C) $Cu + ZnCl_2$** D) $Cl_2 + KBr$

4.5 Double Replacement & Solubility Rules

Swap partners • precipitate, water or gas

BIG IDEA

In a double replacement reaction two compounds exchange partners: AB + CD → AD + CB. In water the reaction only "happens" if one new compound is a **solid (precipitate)**, **water** or a **gas**; otherwise the ions just stay mixed (no reaction).



Two clear solutions form a solid precipitate (colored yellow in real life).

Solubility rules (water)	Soluble (aq) ...	... except
Group 1 ions (Li^+, Na^+, K^+) and NH_4^+	always soluble	none
Nitrates (NO_3^-), acetates ($C_2H_3O_2^-$), chlorates	always soluble	none
Chlorides, bromides, iodides	soluble	Ag^+, Pb^{2+}
Sulfates	soluble	Ba^{2+}, Sr^{2+}, Pb^{2+}, Ca^{2+}(slightly), Ag^+
Hydroxides, carbonates, phosphates, sulfides, sulfites	insoluble (s)	Group 1, NH_4^+ (and Ba^{2+}, Sr^{2+}, Ca^{2+} hydroxides)

STEPS

1. Swap the cations. 2. Write each new formula using ion charges (criss-cross). 3. Balance. 4. Use the solubility rules to label (s) or (aq). 5. If both products are aqueous: **no reaction**.

#	Reactants	Products (balanced, with states) or NR
11	$BaCl_2 + H_3PO_4$ (aq) →	$3BaCl_2$ (aq) + $2H_3PO_4$ (aq) → $Ba_3(PO_4)_2$ (s) + 6HCl (aq)
12	$K_2SO_4 + MgF_2$ (aq) →	NR (no solid, gas or water forms)
13	$AlCl_3 + BaSO_4$ (aq) →	NR (no solid, gas or water forms)
14	$Li_2S + AgNO_3$ (aq) →	Li_2S (aq) + $2AgNO_3$ (aq) → $2LiNO_3$ (aq) + Ag_2S (s)
15	$BaCl_2 + Na_2CO_3$ (aq) →	$BaCl_2$ (aq) + Na_2CO_3 (aq) → $BaCO_3$ (s) + 2NaCl (aq)
16	$Al_2(SO_4)_3 + Na_3PO_4$ (aq) →	$Al_2(SO_4)_3$ (aq) + $2Na_3PO_4$ (aq) → $2AlPO_4$ (s) + $3Na_2SO_4$ (aq)
17	$Na_2SO_4 + NH_4I$ (aq) →	NR (no solid, gas or water forms)
18	$Li_3PO_4 + BaCl_2$ (aq) →	$2Li_3PO_4$ (aq) + $3BaCl_2$ (aq) → 6LiCl (aq) + $Ba_3(PO_4)_2$ (s)

#	Reactants	Products (balanced, with states) or NR
19	$AlBr_3 + H_3PO_4$ (aq) →	$AlBr_3$ (aq) + H_3PO_4 (aq) → $AlPO_4$ (s) + 3HBr (aq)
20	$Zn(NO_3)_2 + Ba(OH)_2$ (aq) →	$Zn(NO_3)_2$ (aq) + $Ba(OH)_2$ (aq) → $Zn(OH)_2$ (s) + $Ba(NO_3)_2$ (aq)

TEACHER NOTE

The homework numbering continues from #11. #13 lists $BaSO_4$ as a reactant (aq); $BaSO_4$ is insoluble, so the key calls it NR. Weak acids/gases (H_2CO_3 → $H_2O + CO_2$, NH_4OH → NH_3) can be mentioned as extra 'driving forces' but are not needed for the exam.

1. Which pair gives a precipitate?

A) NaCl + KNO_3 **B) $Pb(NO_3)_2$ + KI** C) $NH_4Cl + NaNO_3$ D) K_2SO_4 + NaCl

2. What is the precipitate when $AgNO_3$(aq) and NaCl(aq) are mixed?

A) $NaNO_3$ **B) AgCl** C) $AgNO_3$ D) no precipitate

4.6 Net Ionic Equations *Show only the particles that change*

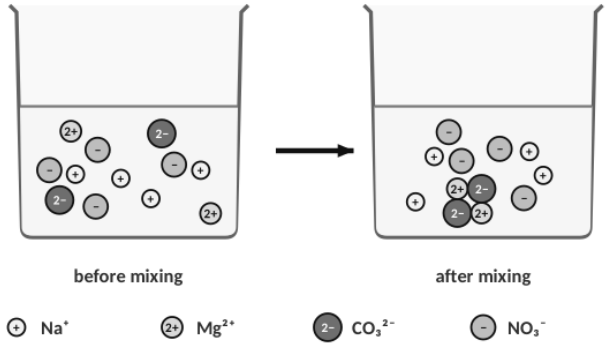
Key terms

Term	Definition (your words)	Example / sketch
Net ionic equation	An equation that shows only the particles that actually change in a reaction; spectator ions are left out.	Ag^+(aq) + Cl^-(aq) → AgCl(s)
Spectator ion	An ion that is present before and after the reaction but doesn't take part (it stays dissolved).	Na^+ and NO_3^- in the example

HOW TO WRITE ONE

1. Balanced molecular equation with states.
2. Split **soluble ionic compounds (aq)** into ions. Keep solids (s), liquids (l), gases (g) and weak electrolytes together.
3. Cancel the ions that appear unchanged on both sides (spectators).
4. Write what is left; check atoms AND charge.

A student dissolves sodium carbonate and magnesium nitrate in water. The diagrams show the beaker before and after the reaction.



1. (a) Write the balanced equation. (b) Which ions react and which are spectators? (c) Justify with the diagram. (d) Write the net ionic equation.

(a) Na_2CO_3 (aq) + $Mg(NO_3)_2$ (aq) → $2NaNO_3$ (aq) + $MgCO_3$ (s)
(b) Reacting: Mg^{2+} and CO_3^{2-}. Spectators: Na^+ and NO_3^-.
(c) In the diagram, the Na^+ and NO_3^- ions are still dissolved and unchanged after the reaction, while the Mg^{2+} and CO_3^{2-} ions are gone from the solution and appear as the solid $MgCO_3$ at the bottom.
(d) CO_3^{2-} (aq) + Mg^{2+} (aq) → $MgCO_3$ (s)

Write the net ionic equation

	Reaction	Net ionic equation
a	$SrBr_2(aq) + K_2SO_4(aq) \rightarrow SrSO_4(s) + 2KBr(aq)$	Sr^{2+} (aq) + SO_4^{2-} (aq) → $SrSO_4$ (s)
b	$3Ca(OH)_2(aq) + 2FeCl_3(aq) \rightarrow 3CaCl_2(aq) + 2Fe(OH)_3(s)$	$3OH^-$ (aq) + Fe^{3+} (aq) → $Fe(OH)_3$ (s)
c	$2AgOH(s) + H_2CO_3(aq) \rightarrow Ag_2CO_3(s) + 2H_2O(l)$	$2AgOH(s) + H_2CO_3(aq) \rightarrow Ag_2CO_3(s) + 2H_2O(l)$ (no spectators: every species is a solid, a liquid or a weak acid, so nothing is split)
d	$Fe(s) + NiCl_3(aq) \rightarrow FeCl_3(aq) + Ni(s)$	$Fe(s) + Ni^{3+}(aq) \rightarrow Fe^{3+}(aq) + Ni(s)$ (Cl^- is the spectator)
e	Sodium sulfide(aq) + iron(II) nitrate(aq) →	Na_2S (aq) + $Fe(NO_3)_2$ (aq) → $2NaNO_3$ (aq) + FeS (s) Net: S^{2-} (aq) + Fe^{2+} (aq) → FeS (s)

TEACHER NOTE

Check (b): $Ca(OH)_2$ is treated as soluble here (the homework equation lists it as aqueous). Net ionic: $Fe^{3+}(aq) + 3OH^-(aq) \rightarrow Fe(OH)_3(s)$. In (d) the charges balance: +3 −3 = 0 on the left, neutral solid on the right. Common errors: splitting the solid, forgetting to reduce the coefficients, losing charge balance.

2. Spectator ions are…

A) ions that form the precipitate
B) ions that stay dissolved and unchanged
C) ions that gain electrons
D) ions in the solid

4.7 Naming & Writing Formulas (Review) *Ionic • molecular • acids*

Type	Rule	Example
Ionic (metal + non-metal)	cation name + anion name (-ide); transition metals get a Roman numeral for the charge	Li_2S = lithium sulfide; $FeCl_3$ = iron(III) chloride
With polyatomic ion	name of the cation + name of the polyatomic ion (no change)	$Mg(C_2H_3O_2)_2$ = magnesium acetate
Molecular (2 non-metals)	prefixes: mono, di, tri, tetra, penta, hexa, hepta, octa; no 'mono' on the first element	N_2O_3 = dinitrogen trioxide
Binary acid	hydro- + root + -ic acid	HCN = hydrocyanic acid; HCl = hydrochloric acid
Oxyacid	-ate ion → -ic acid; -ite ion → -ous acid	$HClO_3$ = chloric acid (from chlorate)

	Given	Name or formula
1	Li_2S	lithium sulfide
2	beryllium phosphide	Be_3P_2
3	$Be(OH)_2$	beryllium hydroxide
4	P_2Br_4	diphosphorus tetrabromide
5	SF_6	sulfur hexafluoride
6	magnesium acetate	$Mg(C_2H_3O_2)_2$
7	diboron tetrahydride	B_2H_4
8	dinitrogen trioxide	N_2O_3
9	hydrocyanic acid	HCN
10	lithium iodide	LiI
11	$KMnO_4$	potassium permanganate
12	$HClO_3$	chloric acid

1. Which is the correct name for N_2O_3?

A) nitrogen oxide **B) dinitrogen trioxide** C) nitrogen(III) oxide D) dinitrogen oxide

PRACTICE

Extra Practice — Unit 4

Show your work.

A. Balance

	Balance each equation (show your atom tally below it)
a	$Al + HCl \rightarrow AlCl_3 + H_2$ ANSWER: $2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$
b	$Fe + O_2 \rightarrow Fe_2O_3$ ANSWER: $4Fe + 3O_2 \rightarrow 2Fe_2O_3$
c	$C_4H_{10} + O_2 \rightarrow CO_2 + H_2O$ ANSWER: $2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$
d	$Ca(OH)_2 + H_3PO_4 \rightarrow Ca_3(PO_4)_2 + H_2O$ ANSWER: $3Ca(OH)_2 + 2H_3PO_4 \rightarrow Ca_3(PO_4)_2 + 6H_2O$

B. Classify and predict

Reactants	Balanced products	Type
$Mg + O_2$	$2Mg + O_2 \rightarrow 2MgO$	synthesis
HgO (heat)	$2HgO \rightarrow 2Hg + O_2$	decomposition
$C_3H_8 + O_2$	$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$	combustion
Zn + HCl	$Zn + 2HCl \rightarrow ZnCl_2 + H_2$	single replacement

Reactants	Balanced products	Type
$AgNO_3 + CaCl_2$	$2AgNO_3 + CaCl_2 \rightarrow 2AgCl + Ca(NO_3)_2$	double replacement
$HCl + Ca(OH)_2$	$2HCl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O$	neutralization

C. Precipitation lab data

A student mixes pairs of 0.1 M solutions and records whether a solid forms. Predict the result and write the net ionic equation for the ones that react.

Solutions mixed	Precipitate?	Net ionic equation
$NaCl + AgNO_3$	yes	Cl^- (aq) + Ag^+ (aq) → AgCl (s)
$KI + Pb(NO_3)_2$	yes	$2I^-$ (aq) + Pb^{2+} (aq) → PbI_2 (s)
$NaOH + CuSO_4$	yes	$2OH^-$ (aq) + Cu^{2+} (aq) → $Cu(OH)_2$ (s)
$KNO_3 + NaCl$	no	no reaction
$BaCl_2 + K_2SO_4$	yes	Ba^{2+} (aq) + SO_4^{2-} (aq) → $BaSO_4$ (s)

D. Think

1. A student balances $H_2 + O_2 \rightarrow H_2O$ by changing it to $H_2 + O_2 \rightarrow H_2O_2$. Explain what is wrong with this method.

Changing a subscript changes the substance (hydrogen peroxide is not water); only coefficients may be changed: $2H_2 + O_2 \rightarrow 2H_2O$.

2. Explain why no reaction occurs when NaCl(aq) and KNO_3(aq) are mixed.

All four ions (Na^+, Cl^-, K^+, NO_3^-) form soluble compounds; no solid, gas or water forms, so the ions just stay mixed. All ions are spectators.

3. Which is a correct net ionic equation for $AgNO_3 + NaCl$?

A) $AgNO_3 \rightarrow Ag^+ + NO_3^-$ **B) $Ag^+ + Cl^- \rightarrow AgCl(s)$** C) $Na^+ + NO_3^- \rightarrow NaNO_3$ D) $Ag + Cl \rightarrow AgCl$

4. Which reaction type is: $2Mg + O_2 \rightarrow 2MgO$?

A) decomposition **B) synthesis** C) single replacement D) double replacement

Unit 4 review

5. Describe the steps for predicting the products of a reaction in each of the six reaction types.

Synthesis: join the elements (use charges). Decomposition: split a binary compound into its elements. Neutralization: acid + base → salt + water. Combustion: $CO_2 + H_2O$. Single replacement: use the activity series (more active replaces less active). Double replacement: swap partners; check for precipitate with solubility rules; then balance.

6. Explain the difference between a balanced equation, a complete ionic equation and a net ionic equation.

Balanced: all species as formulas. Complete ionic: aqueous strong electrolytes written as ions. Net ionic: spectators removed, leaving only the particles that change.

Sources & further reading

- PhET – Balancing Chemical Equations simulation (University of Colorado Boulder) — https://phet.colorado.edu/en/simulations/balancing-chemical-equations
- LibreTexts – 11.08 Activity Series (CK-12) — https://chem.libretexts.org/Bookshelves/Introductory_Chemistry/Introductory_Chemistry_(CK-12)/11:_Chemical_Reactions/11.08:_Activity_Series
- LibreTexts – Types of Chemical Reactions: Single and Double Replacement (Ball) — https://chem.libretexts.org/Bookshelves/Introductory_Chemistry/Beginning_Chemistry_(Ball)/04%3A_Chemical_Reactions_and_Equations/4.03%3A_Types_of_Chemical_Reactions_-_Single_and_Double_Replacement_Reactions
- LibreTexts – Single Replacement Reactions and Batteries (lab) — https://chem.libretexts.org/Ancillary_Materials/Laboratory_Experiments/Wet_Lab_Experiments/General_Chemistry_Labs/Online_Chemistry_Lab_Manual/Chem9_Experiments/09:_Single_Replacement_Reactions_and_Batteries_(Experiment)